

Design for Environment Assessment

Objective

To apply lifecycle thinking and the four primary Design for Environment concepts to ideate design improvements and identify indicators/metrics that could feasibly be calculated to quantify/qualify the improvements for your group's four products or processes from the project brief.

Tasks

In your groups, each student will take responsibility for one product / process, identified in the project brief, to analyse using the Design for Environment lens. The analysis of each product / process should include the following subsections:

1. Systems and Life Cycle thinking
2. Design for Dematerialization
3. Design for Detoxification
4. Design for Revalorization
5. Design for Capital Protection and Renewal

For each of these areas, you will typically (i) describe how they are relevant or irrelevant to your product / process, (ii) develop ideas how the product/process could be better designed to follow these principles, and (iii) describe Performance Indicators and Metrics that could be used to evaluate the ideas. See the marking rubric to understand how the report will be graded.

Report

The group report must include the following sections:

- Title page
- Executive summary (one page max)
- 1. Introduction (one page max)
- 2. Student A's analysis* (four pages max)
- 3. Student B's analysis* (four pages max)
- 4. Student C's analysis* (four pages max)
- 5. Student D's analysis* (four pages max)
- 6. Discussion (two pages max)
- 7. Conclusion (One page max)
- 8. References

* Students should give an appropriate title to their section followed by their name in brackets, e.g. Concrete Bridge (Joe Smith).

Student Contributions to Group Sections

There is no need to indicate the writer of the group sections. It is up to the groups to fairly distribute the workload. Instructions will be given via Moodle for cases where this is clearly not the case.

Grading

The marker will use the marking rubric to assess the standard and coverage of your work. The report grade will be 80% individual and 20% group.

Plagiarism

The university has a strict plagiarism policy. Be sure to follow the policy. Please do not search the internet for an existing LCA study, copy its topic, and then re-write the study. Be honest.

Late Submission

Late work is penalised by reducing the grade by 5% per working day, down to a minimum of 50%. No mark will be awarded if the assessment item misses the final cut-off date, which is 1 week after the due date in the schedule. If you wish to seek Special Consideration, see Moodle for instructions and contact the paper convenor before the due date of the assessment.